

LWAZI KNOWLEDGE GUMEDE

ECSA Candidate Engineer | BSc (Honours) Computer Engineering | Full-Stack Developer

lwazig28@gmail.com • +27 76 935 2103 / +27 65 711 1226 • Umlazi, Durban, South Africa • Web • GitHub

PROFESSIONAL SUMMARY

ECSA-registered Candidate Engineer and BSc (Honours) Computer Engineering graduate who builds and ships full-stack software inside a live rail-infrastructure environment. Combines hands-on field engineering, including CAS testing, commissioning, and derailment investigations, with production software delivery in Python, Django, React, TypeScript, and PostgreSQL. Strong in system automation, data engineering, and REST API development, with a consistent record of replacing manual workflows with reliable automated tools, and just as comfortable across hardware and software through embedded and IoT systems.

WORK EXPERIENCE

Engineer-in-Training

April 2025 - Present

Transnet SOC Ltd (Rail Infrastructure Manager) | Johannesburg

- Replaced manually compiled spreadsheets, saving the team the equivalent of roughly a week of manual work each month, by designing and deploying full-stack monitoring dashboards in React, TypeScript, Python/Django, PostgreSQL, and WebSocket that gave the Technical Office and corridor depots a single live view of rail operations.
- Removed repetitive manual logging from daily engineering workflows by building automated Python/REST backend services for data collection, vehicle identification, and compliance reporting.
- Strengthened asset-reliability and safety decisions by running technical reviews, field testing, and commissioning of Condition Assessment Systems (CAS) and supporting derailment investigations with documented analysis and configuration management.
- Stayed on track for professional registration by logging engineering competencies against ECSA candidate requirements in a documented professional-development logbook.

Engineering Tutor & Laboratory Demonstrator

February 2024 - May 2025

University of KwaZulu-Natal | 5 modules, first- and second-year engineering

- Raised the standard of student technical reports and presentations by running Technical Communication (ENCH1TC) tutorials and giving structured written feedback, earning consistently positive lecturer and student ratings for clarity.
- Helped students turn theory into working solutions by demonstrating Computer Methods (programming & numerical analysis), Electronic Design, Electrical Principles (ENEL2EB), and Field Theory (ENEL2FT) labs and guiding projects from concept to implementation.
- Ensured accurate experimental results and full safety compliance by setting up, verifying, and supervising practical lab equipment for second-year cohorts.

Vacation Student (Engineering Internship)

January 2023 - March 2023

Transnet Engineering | Durban

- Gave engineering stakeholders self-serve access to operational data by building Power BI dashboards that turned raw engineering metrics into clear visual reports for decision-making.
- Eliminated manual data hand-offs by implementing Python-scripted pipelines that automatically pushed data into shared Google Sheets for real-time, cross-department access.
- Built early exposure to rail operations, engineering project management, and safety protocols within a large state-owned enterprise.

EDUCATION

BSc (Honours) in Computer Engineering

2019 - June 2024

University of KwaZulu-Natal

Focus areas: Embedded Systems, IoT, Software Development, Digital Signal Processing, Control Systems, Data Structures and Algorithms, Database Systems. Key projects: IoT Smart Meter System, Blockchain Invoice Verification, Multi-Factor Authentication System, PIC16 Embedded Game System, JPEG Image Compression.

National Senior Certificate (NSC / Matric)

2018

Nombuso High School | Bachelor's Degree Pass

Subjects: Mathematics, Physical Sciences, Life Sciences, Agricultural Science, English FAL, isiZulu Home Language, Life Orientation

SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, PHP, SQL, Solidity, Bash, Assembly Language, MATLAB

Web Development: React, Django, Flask, Node.js, Next.js, Express.js, HTML5, CSS3, Bootstrap, RESTful APIs, WebSocket, D3.js

Databases: MySQL, PostgreSQL, SQLite, MongoDB, Database Design and Optimisation

Tools and Technologies: Git, GitHub, Docker, Linux, Visual Studio Code, Power BI, Arduino, Microcontrollers, Postman

Engineering Tools: MATLAB, Simulink, Proteus, Altium Designer, Oscilloscope

Specialised Skills: IoT Systems, Blockchain Development, Embedded Systems, System Automation, Microcontroller Programming, CI/CD Pipelines

Methodologies: Agile, Scrum, Software Testing and Quality Assurance, Technical Documentation

Soft Skills: Problem-solving, Cross-functional collaboration, Technical communication, Research, Adaptability, Safety awareness

Languages: English (Professional), Zulu (Native), Xhosa (Conversational)

CERTIFICATIONS & PROFESSIONAL REGISTRATION

ECSA Candidate Engineer - Engineering Council of South Africa (ECSA) | Registered 2025

Engineer-in-Training (EIT) Programme - ECSA-accredited 2-year structured training, Transnet SOC Ltd | April 2025 - March 2027

BSc (Honours) Computer Engineering - ECSA-accredited degree, University of KwaZulu-Natal | Graduated June 2024

REFERENCES

Lungelihle Jafta - Line Manager, Transnet SOC Ltd (Rail Infrastructure Manager)

Dr. Jules-Raymond Tapamo - Lecturer, UKZN (Computer Methods)

Wayne Nelson - Lecturer, UKZN (Technical Communication)

Adv. Dr. Ernest Bhero - Lecturer, UKZN (Electronic Engineering Design)